What Is PeX?

A structured Payment Exception Resolution intelligence layer — built around 41 exception profiles.

1. THE 41 EXCEPTION UNIVERSE · *Canonical Scope*

- **Taxonomy Scope:** Spans 41 BIAN-aligned canonical payment exception profiles across 5 operational domains:
- **Account & Routing (12):** Invalid IBAN/BBAN, closed accounts, missing correspondent BIC, routing table mismatch.
- **Data & Format (9):** ISO 20022 XML syntax errors, MT103 field truncation, charge code (OUR/BEN/SHA) errors.
- **Compliance & Sanctions (8):** OFAC 50% disambiguation, PEP false-positive elimination, FATF Travel Rule gaps.
- **Liquidity & Timing (6):** Clearing cutoff breaches, debit cap limits, value-date mismatch, cover delays.
- **Duplicate & Processing (6):** Duplicate UETR tracking, pacs.004 return processing, camt.056 recall handling.

2. WHAT PeX ADDS · *Logical Capabilities*

- **Exception Intelligence:** Ingests raw alerts, parses multi-rail payloads, and derives the root-cause exception profile.
- **Evidence & Context:** Asynchronously queries peripheral bank systems (DDA, Sanctions, Client Master) via read-only ICD adapters.
- **Hypothesis Elimination:** Evaluates and mathematically scores competing resolution hypotheses.
- **Deterministic Rules:** Applies 100% authoritative banking rules before any proposal is submitted.
- **Human Authorization:** Enforces strict Maker-Checker Segregation of Duties ($0 threshold) and Dual-Auth (≥$250k).
- **Merkle Provenance:** Cryptographically seals every decision node into an immutable SHA-256 chain.

3. WHY THIS IS MORE THAN AI · *Enterprise Distinction*

- **Reasoning Scope:** Generic LLMs generate unconstrained text; PeX produces typed, schema-validated ReasoningResult JSON.
- **Domain Centricity:** Generic AI is prompt-centric; PeX is grounded in 41 Canonical Exception Profiles.
- **Governance & Safety:** Generic AI hallucinates; PeX binds AI advice with 100% authoritative deterministic rules.
- **Data Context:** Generic AI uses static prompts; PeX pulls live corroborating evidence via secure bank ICDs.
- **Auditability:** Generic AI offers opaque logs; PeX provides an unbroken cryptographic SHA-256 audit trail.

CANONICAL RESOLUTION LAYER ► 41 EXCEPTION PROFILES ► BANKING ONTOLOGY ► CANONICAL MODEL ► PERIPHERAL EVIDENCE ► COMMON RESOLUTION INTELLIGENCE

CORE TAKEAWAY: PeX is not 'AI for payments.' It is a structured resolution capability built around a defined exception universe.